FSWD LAB 6

MARYAM JAHANGIR 241949

testing pro > index.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Testing Project</title>
</head>
<body>
    <script src="demo.js"></script>
</body>
</html>
```

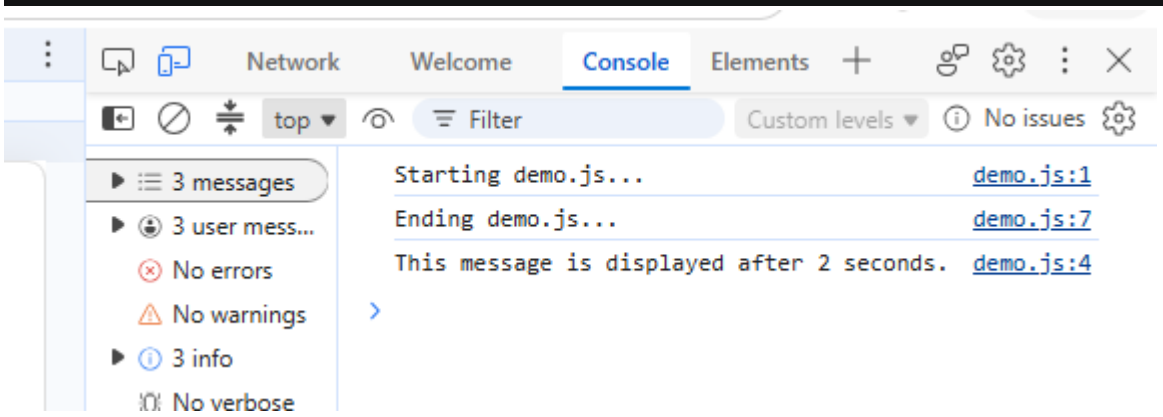
Lab5 > testing pro > JS demo.js > ...

```js
console.log("Starting demo.js...");

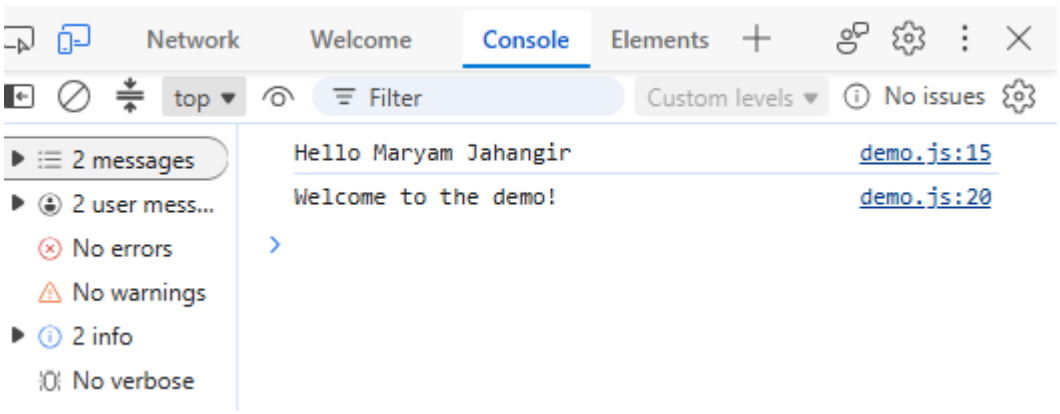
setTimeout(() => {
    console.log("This message is displayed after 2 seconds.");
}, 2000);

console.log("Ending demo.js...");

```



```
//call back function  Synchronous
// The main function that accept function as argument
function greet(name, welcome){  /* the first one is paramerter and second one is
    call back function name that did not have paranthesis because we are not calling
    it here we are just passing it as a reference to the main function
    and it's not an error to have different names */
    console.log("Hello " + name);
    welcome();  // calling the call back function
}
// call back function definition
function welcomeMessage(){
    console.log("Welcome to the demo!");
}
// calling the main function and passing the call back function as an argument
greet("Maryam Jahangir", welcomeMessage);
```

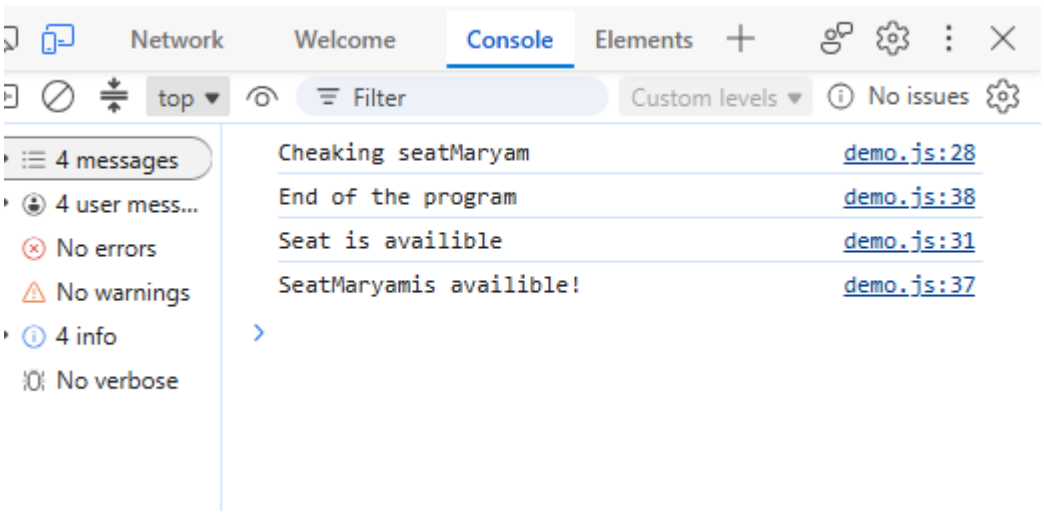


```
// Call back fuction Asynchronous
function checkseat(seatnumber, callback){
    console.log("Cheaking seat" + seatnumber);

    setTimeout(()=>{
        console.log("Seat is availible");

        callback("Seat" + seatnumber + "is availible!")
    }, 2000);
}

checkseat("Maryam", (message)=>{console.log(message)});
console.log("End of the program");
```

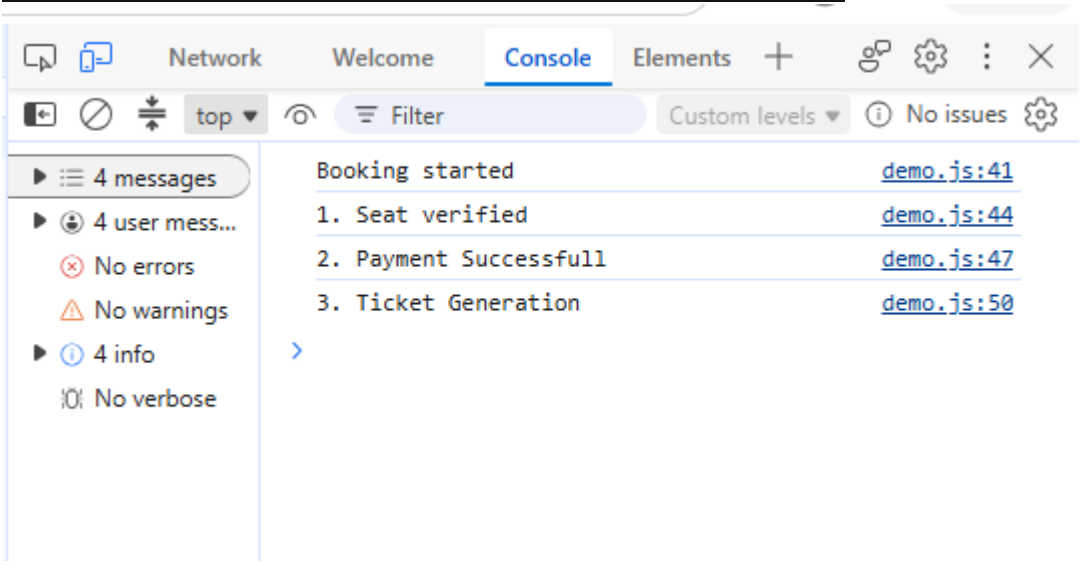


```
// Call Back Hell
console.log("Booking started");

setTimeout(()=>{
    console.log("1. Seat verified");

    setTimeout(()=>{
     console.log("2. Payment Successfull")

     setTimeout(()=>{
        console.log("3. Ticket Generation");
     }, 2000);
    }, 5000);
}, 5000);
```



```
// Call Back Hell
console.log("Booking started");

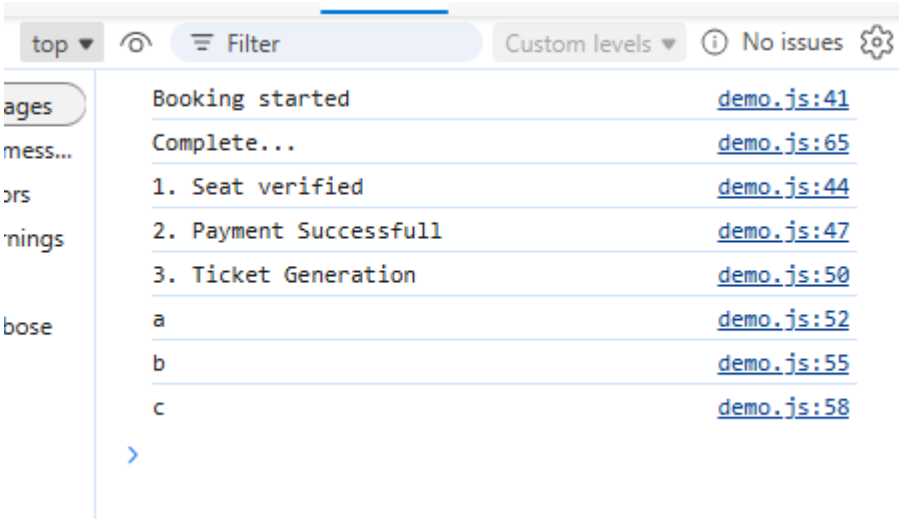
setTimeout(()=>{
    console.log("1. Seat verified");

    setTimeout(()=>{
     console.log("2. Payment Successfull")

     setTimeout(()=>{
        console.log("3. Ticket Generation");
    setTimeout(() => {
        console.log("a");

    setTimeout(()=>{
        console.log("b");

    setTimeout(()=>{
        console.log("c");
    }, 2000);
    },2000)
    }, 2000);
     }, 2000);
    }, 5000);
}, 5000);
console.log("Complete...");
```

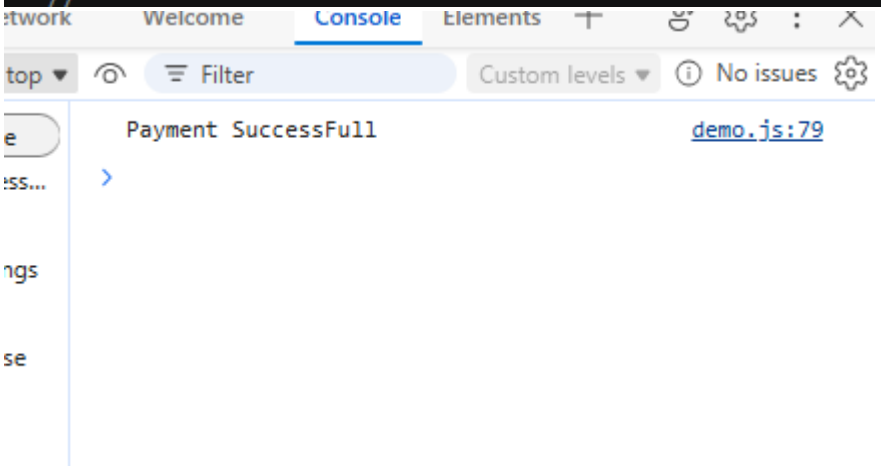


```
// console.log("Complete...");

// Promise
const payment = new Promise((resolve, reject)=>{
  let success = true;
if(success){
    resolve("Payment SuccessFull")
}
else {
    reject(new Error("Payemnt failed!"));
}

});
payment.then((message)=>{
    console.log(message);
});
payment.catch((error)=>{
    console.log(error);
});

//
```

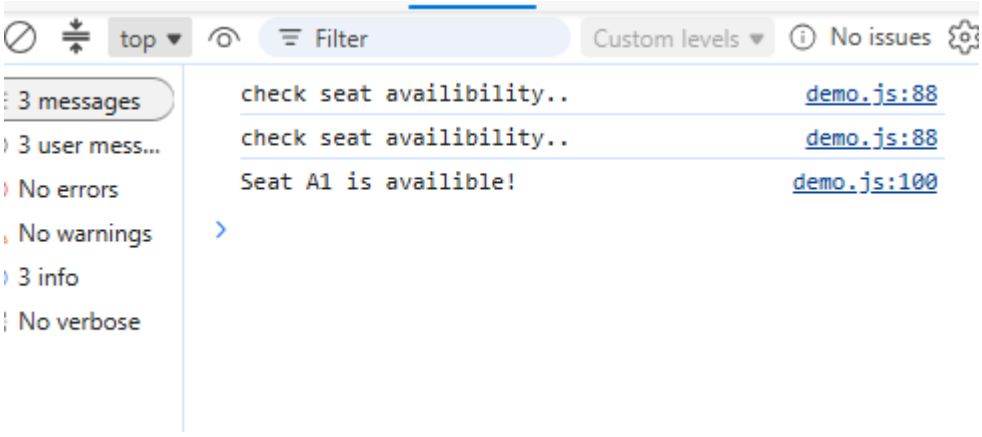


```
//
function verifyseat(){
    return new Promise((resolve, reject)=>{
      console.log("check seat availibility..");

      setTimeout(()=>{
        resolve("Seat A1 is availible!")
      }, 2000);
       setTimeout(()=>{
        resolve("Seat A1 is Not Available!")
      }, 2000);
    });
}

verifyseat().then(()=>{
   console.log("Seat A1 is availible!");
})
.catch((error)=>{
    console.log(error.message);
})

verifyseat(()=>{    // same thing with callback hell
    processPayment(()=>{
        issueTicket(()=>{
            console.log("Ticket issued successfully!");
        });
});
});
```



```
function verifyseat(){
    return new Promise((resolve, reject)=>{
      console.log("check seat availibility..");

      setTimeout(()=>{
        resolve("Seat A1 is availible!")
      }, 2000);
       setTimeout(()=>{
        resolve("Seat A1 is Not Available!")
      }, 2000);
    });
}
verifyseat().then(()=>{
   console.log("Seat A1 is availible!");
})
.catch((error)=>{
    console.log(error.message);
})

// verifyseat(()=>{    // same thing with callback hell
//     processPayment(()=>{
//         issueTicket(()=>{
//             console.log("Ticket issued successfully!");
//         });
// });
// });
verifyseat()    // same thing with Promise
  .then(() => processPaymentPromise())
  .then(() => issueTicketPromise())
  .catch((error) => {
    console.error("Error:", error.message);
  });
```

```
async function bookTicket() {
    await verifyseat();
    await processPaymentPromise();
    await issueTicketPromise();
    console.log("Ticket booked successfully!");
  }
bookTicket();
```